Kavin Gunasekaran

4 Cumberland St, Brantford, Ontario N3S7M5

226-998-3547 • kgunasek@uoguelph.ca • https://www.linkedin.com/in/kavin-gunasekaran-201435252/ • https://github.com/KavinG1242 • Research Papers Projects

Technical & Communication Skills

- IDE: Visual Studio, Unity, Netbeans, Vivado, MatLab, Arduino, R studio, Keil, STM32CubeIDE
- Programming Languages: C, Java, JavaFX, C#, VHDL, Verilog, ARM, MIPS, Python, MatLab, R, LaTeX, C++, HTML
- Software Experience: Adobe Creative Suite, MS Office Suite, Unity Game Engine, Google Works, Solidworks, AutoCad, KiCAD, Cisco DNAC, NetBox, Cisco Unified Communications Messenger, Cisco Prime Infrastructure, Cisco Unity, Intel Quartus, Jenkins
- Operating Systems: Windows, Macintosh, Linux

Education

University of Guelph, *Bachelor of Engineering, Computer Engineering (Coop).*
Guelph, Ontario • September 2021 - Present

Flow-Induced Vibration Testing for Small Modular Reactors (SMRs) September 2025 - Present
University of Guelph Capstone Project

- Developing an experimental platform to analyze two-phase flow-induced vibrations in Small Modular Reactor (SMR) heat-exchanger tubing.
- Implementing a PID-controlled system in LabVIEW and STM32CubeIDE to regulate the void fraction of a refrigerant-based working fluid.
- Designing instrumentation and control circuitry in accordance with CNSC Regulatory Guides G-320 & G-278 and CSA N299/B52 standards for nuclear and refrigeration systems
- Aiming to create a scalable testing method that supports the Canadian SMR Roadmap by improving vibration monitoring, safety, and modular testing efficiency.

Power-Assisted Gurney for Paramedic Ergonomics January 2025 - April 2025
University of Guelph Design III Project

- Conducted a theoretical and simulation-based design of a motor-assisted gurney system to reduce musculoskeletal strain on paramedics during patient transport
- Modelled motor torque, battery capacity, and load-cell sensor feedback using MATLAB, SolidWorks, and KiCAD circuit simulations.
- Designed a microcontroller-driven control scheme (STM32-based) with throttle logic, safety overrides, and power-management algorithms in simulation.
- Evaluated ergonomic, cost, and lifecycle performance metrics; validated the design against CSA D500:20 standards for paramedic transport equipment.

Post-Quantum Cryptography Research January 2025 - April 2025
University of Guelph Embedded Design Project

- Authored a technical paper analyzing quantum computing threats to classical cryptography and surveying post-quantum cryptographic (PQC) algorithms and Quantum Key Distribution (QKD) systems.
- Compared lattice-based, code-based, and hash-based PQC schemes for integration with classical RSA/ECC systems.
- Proposed a hybrid quantum-safe communication model combining PQC algorithms with QKD network protocols for long-term data security.

Professor Availability System September 2024 - December 2024
University of Guelph Personal Project

- Designed and implemented a Professor Availability System using ESP32 nodes for one-way wireless communication, enhancing accessibility for students.
- Developed a web server interface to display availability statuses and stream live video from an ESP32 camera for real-time monitoring.
- Integrated multiple hardware components, including motion sensors, LCD displays, and LEDs, to provide an intuitive and energy-efficient system.
- Resolved hardware and software challenges, such as optimizing voltage supply with step-up modules and debugging LCD display functionality, ensuring reliable performance.

Work Experience

Co-op Software QA Developer May 2025 – December 2025
Per Vices, Toronto, Ontario, Canada.

- Collaborated with development teams to design and maintain automated test suites, implementing new test cases and bug-regression coverage for core software modules.
- Introduced refactoring and optimizations in legacy code paths based on commit-driven improvements, reducing technical debt and improving maintainability.
- Enhanced logging, configuration management, and error handling across multiple modules to facilitate debugging and operational reliability.
- Updated CI/CD build pipelines and configuration files to streamline automated verification, reduce false positives, and accelerate release cycles.
- Authored and maintained documentation, unit tests, and integration tests, aiding onboarding and cross-team collaboration.

Co-op Network Services Technician January 2024 – August 2024
University of Guelph, Computing & Communications Services, Guelph, Ontario, Canada.

- Conducted comprehensive WiFi surveys across campus to optimize network coverage, particularly in high-density areas.
- Collaborated with network administrators to troubleshoot and repair network infrastructure, including desk phones, access points, and UPS systems.
- Gained hands-on experience with Cisco software, efficiently diagnosing and resolving client issues and managing network configurations.
- Played a key role in migrating network databases to NetBox, ensuring a centralized and reliable network source of truth.

Engineering Involvement

Robotics Team Electrical September 2023 - Present
Club

- Designed and developed intricate PCB layouts using KiCAD, ensuring efficient and reliable electronic circuitry for the rover's control systems.
- Collaborated with the electrical team to solder, assemble, and test the PCBs.
- Currently transitioning to firmware development, focusing on programming microcontrollers and optimizing communication between hardware components.

Guelph Engineering Society IT Commisioner September 2023 - Present
Student Leadership

- Updated and enhanced the design of the existing Shopify store to improve user experience and visual appeal.
- Collaborated with team members to revamp the Guelph Engineering Society website, ensuring optimal performance and an updated layout.
- Managed weekly Outlook emails to keep students informed about upcoming events and announcements.
- Assisted in managing the Engineering Equipment Library website built on myturn.com, ensuring smooth operations and up-to-date content.